TUGAS 1 PENGOLAHAN CITRA DIGITAL

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1. Syntax

Tugas1PCD.py

C:\Users\acer\Downloads\TUGAS TUGAS FHAIZ\belajar belajar\Tugas1PCD.py

```python
def matmul_3d(A, B):

    batch_size = len(A)
    m = len(A[0])
    n = len(A[0][0])
    p = len(B[0][0])

    if len(B) != batch_size:
        raise ValueError("Jumlah batch A dan B harus sama")
    if len(B[0]) != n:
        raise ValueError("Kolom A harus sama dengan baris B")

    hasil = []
    for b in range(batch_size):
        matrix_hasil = []
        for i in range(m):
            baris_hasil = []
            for j in range(p):
                total = 0
                for k in range(n):
                    total += A[b][i][k] * B[b][k][j]
                baris_hasil.append(total)
            matrix_hasil.append(baris_hasil)
        hasil.append(matrix_hasil)

    return hasil

A = [
    [[0, 9], [0, 8]],
    [[2, 0], [0, 6]]
]

B = [
    [[4, 3], [2, 1]],
    [[1, 2], [3, 4]]
]

hasil = matmul_3d(A, B)

for idx, m in enumerate(hasil):
    print(f"Hasil batch ke-{idx}:")
    for baris in m:
        print(baris)
    print()
```

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2. Output

```
OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\acer\Downloads\TUGAS TUGAS FHAIZ\belajar belajar> python -u "c:\Users\acer\Downloads\TUGAS TUGAS FHAIZ\belajar belajar\Tugas1PCD
.py"
Hasil batch ke-0:
[18, 9]
[16, 8]

Hasil batch ke-1:
[2, 4]
[18, 24]

PS C:\Users\acer\Downloads\TUGAS TUGAS FHAIZ\belajar belajar>
```